

Boussard & Gavaudan

Convertible Bonds – Teach In

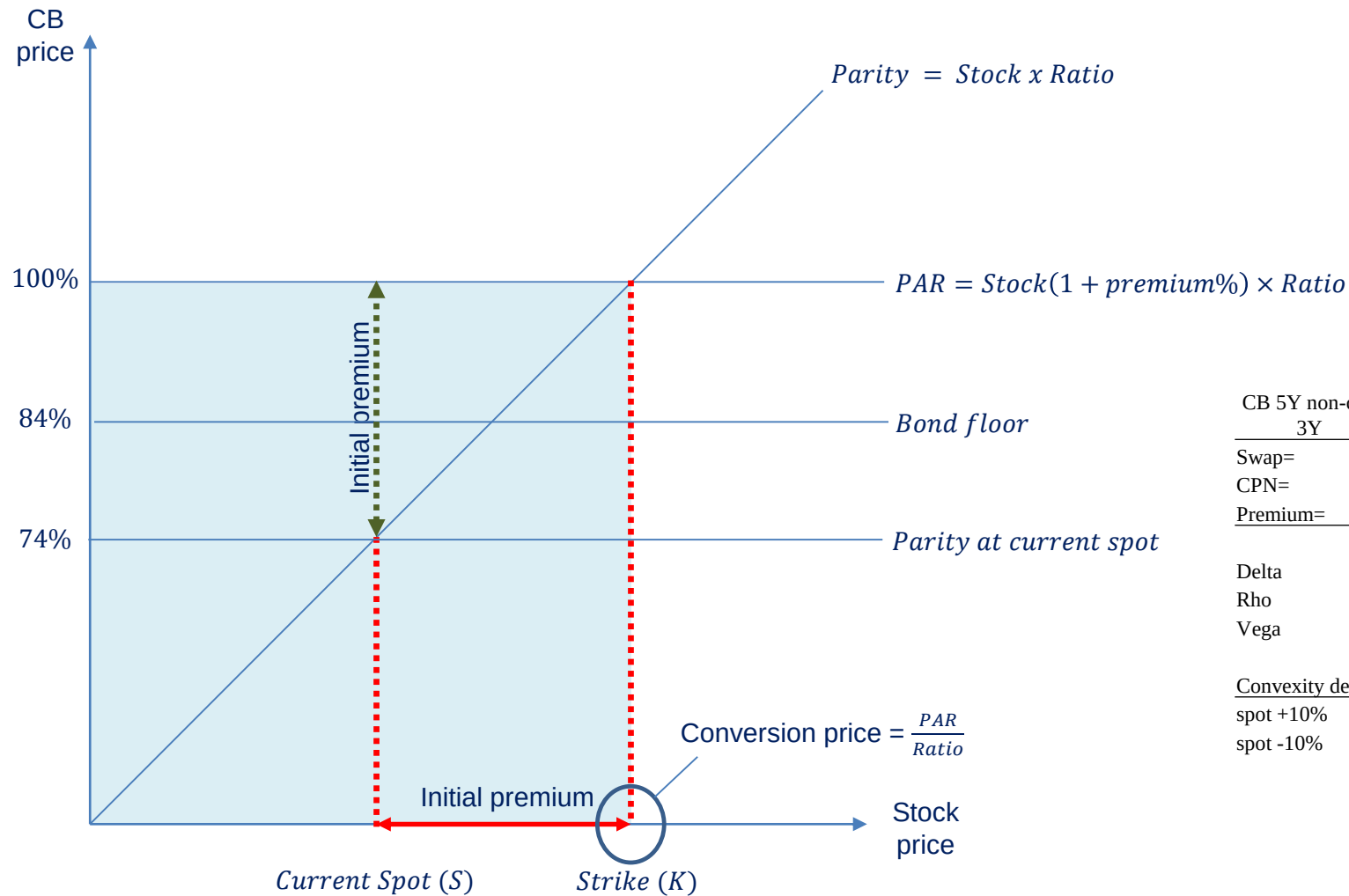
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Convertible Bonds

EXPLANATION			IMPLICATION APPLICATION			
WHAT	WHY ISSUE	WHO BUY	EXPOSURES	STRATEGIES	PARAMETERS	HOW MAKE MONEY
Corporate debt that bears a fixed coupon with a fixed maturity date BOND FLOOR	The option embedded in the CB enables the issuer to pay a lower coupon that would be required on a vanilla bond	Option value decreases every day due to the time decay (theta) Nevertheless, Investor is paid for the liquidity provided (funding interest rate) and the default risk taken (credit spread)	- Rho : the underlying change in price of the security for a unit percentage change in risk-free rate - Xi : same but for credit spread	Long CB outright	- Nominal - Coupon - Yield	Spread => Implied Bond floor
A bond that can be converted into a predetermined number of shares of a specified stock PARITY / PREMIUM	At issuance, Issuer sells Equity with a premium not a discount CB placement process is faster than a capital increase	Long-term maturity option not liquid in the market (if even available)	- Delta: the change in price of the security for a unit change in the underlying share price - Gamma (convexity) - Vega: the change in price of the security for a unit change in volatility	- Arbitrage CB vs Stock - Long only Delta	- Conversion ratio - Conversion price - Premium %	- Spot & Div & Risk Free rate => implied Forward & Parity - Forward & option theory => implied embedded option value of CB (implied premium %)
Debt that offers implicit (from Statute) or explicit (from its prospectus) protections to Investors and options to the Issuers BOND FLOOR ADJUSTMENT PARITY ADJUSTMENT DISTANCE RECALL	Recall features permit issuer with a certain level of stock and non-call period, to "force" bondholders to convert	"Put" features permit bondholders to request early redemption "Ratchet" features provide bondholder with increased conversion rights in the event of a takeover bid to compensate for increased uncertainty	- Recall risk - Dividends - Gain on takeover event	- Arbitrage CB vs Stock - Long only Delta	- Dividend protection - Adjusted conversion price - Ratchet clause - Recall date - Trigger	- Modified bond floor - Adjusted parity & premium

Convertible Bonds – At Issuance



CB 5Y non-call
3Y

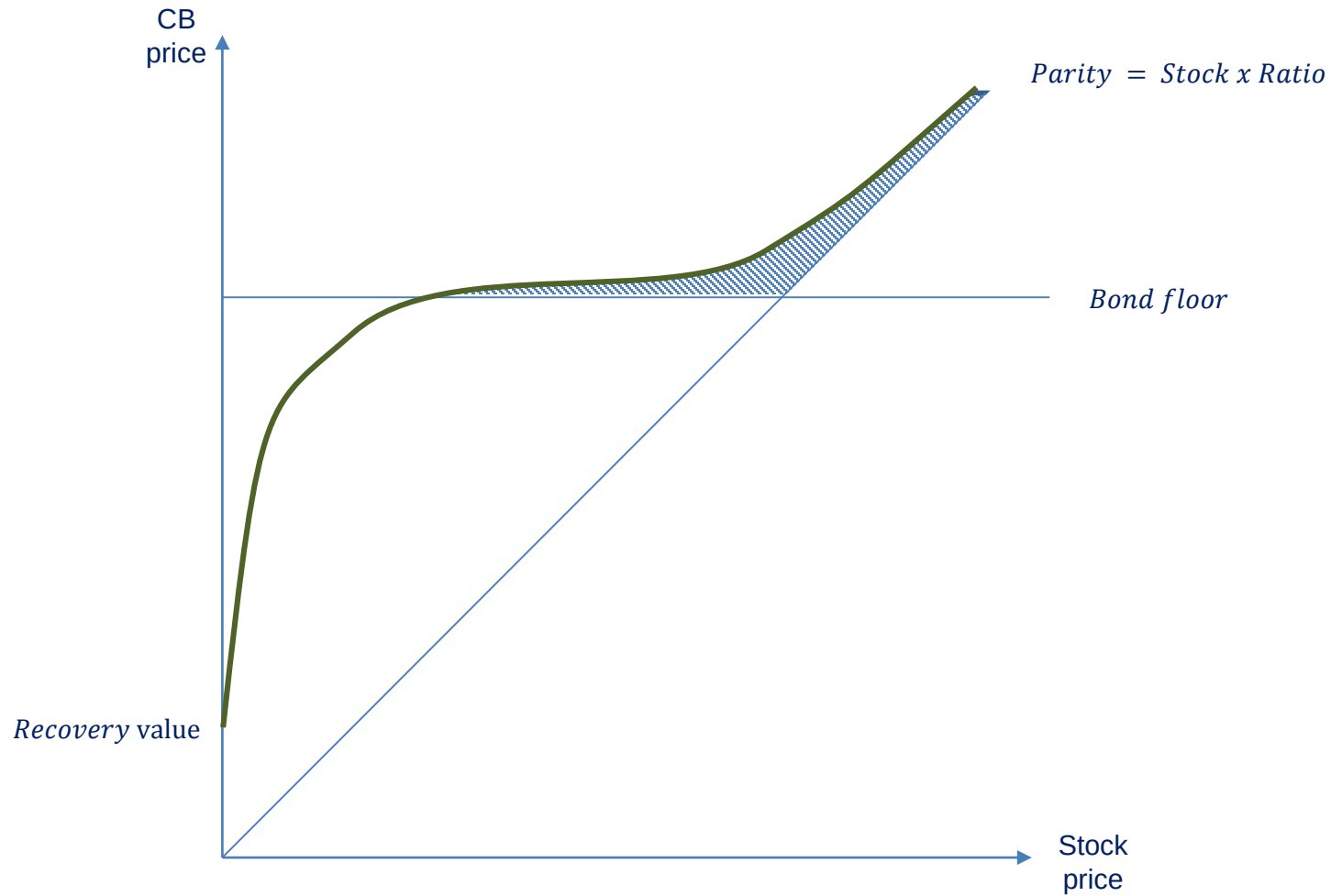
Swap=	0.50%
CPN=	0.50%
Premium=	35%

Delta	0.4pts for 1% change
Rho	2.97pts for 100bps
Vega	0.6pts for 1 vol pt

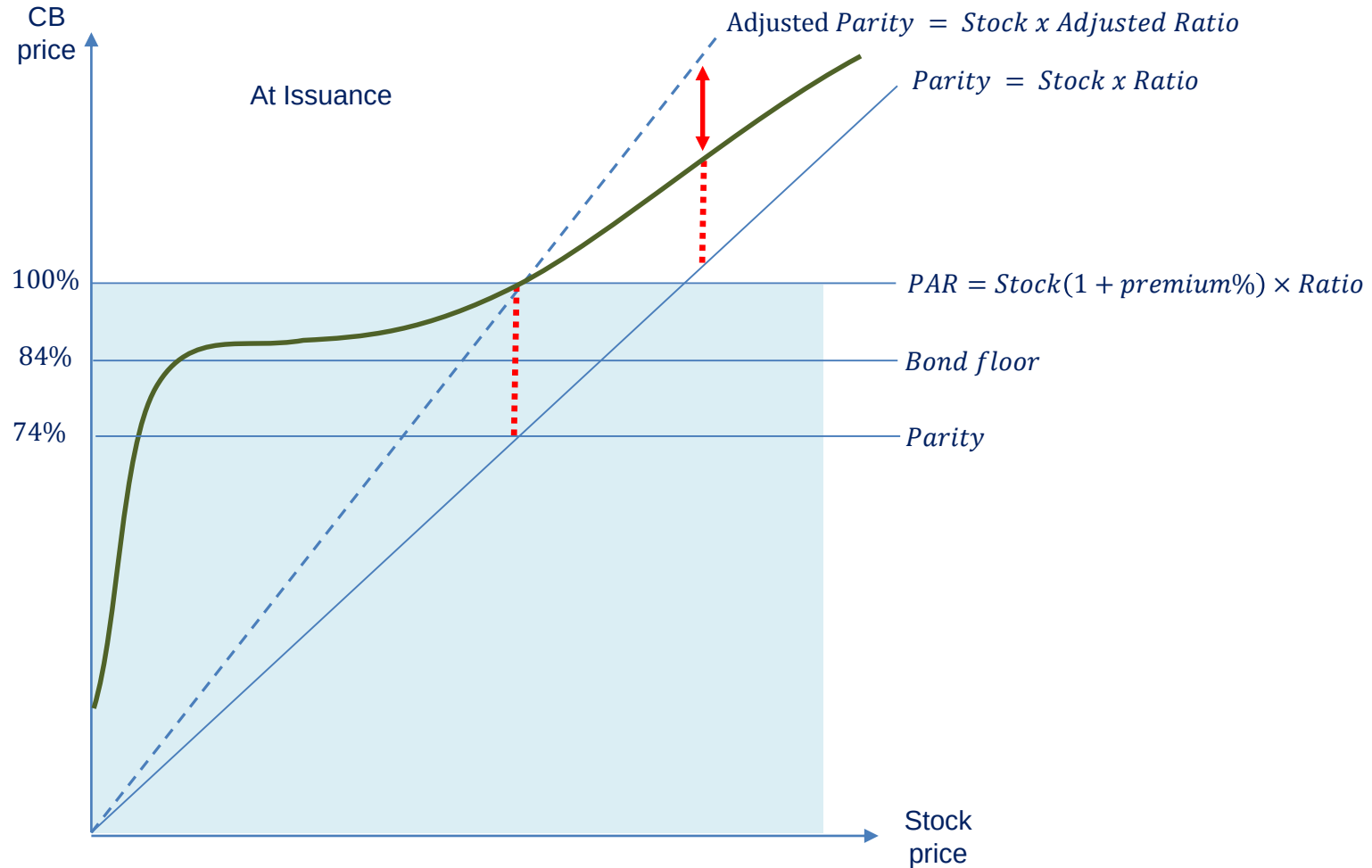
Convexity delta

spot +10%	0.44
spot -10%	0.36

Convertible Bonds – During life-time



Convertible Bonds – Parity adjustment



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